150 A Method for Classifying Large Numbers of End-Joining Events

William Engels
Carlos Flores

University of Wisconsin
Madison
wrengels@wisc.edu

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Abstract

We developed an algorithm and software (*NJoiner*) for analyzing and classifying large collections of NHEJ events. Settings for the method were fine-tuned and optimized by applying it to a collection of 1337 NHEJ events sequenced in our lab plus other data collected from published works. The approach can be used to determine the effect of biological variables (*e.g.,* genetic background, age of organism, genomic position of break) on the features of the NHEJ mechanism. Examples of testable features are (1) frequency and length of microhomology, (2) occurrence of imperfect microhomology, (3) size and frequency of deletions, (4) addition of filler bases and whether such filler is derived from a nearby template. The main advantage of *NJoiner* as opposed to manual classification is that it provides objective and repeatable results that can be compared across experiments. It also eliminates a particularly error-prone, tedious and labor-consuming step in the analysis.

Background

Non-homologous end-joining is a catch-all term for several kinds of double-strand break repair events. These events can be classified according to whether they involve deletions, insertions, microhomology, etc.

We are interested in classifying NHEJ events to determine the effect of genetic background, age of the organism, genomic position of the break, etc.

Large numbers of NHEJ products are not hard to obtain. Our method[3-5] is to use a reporter system called *Rr3* in the Drosophila male germ line. Breaks are made by I-*Sce*I and the repaired products are recovered in the next generation for sequencing.

The Problem

How to classify a given NHEJ event based only on the sequenced product is not always obvious. For example, suppose a DSB is made like this:

CCC…AAGCTGTCCTTCCCCGGCTAGGG↑***ATAA***↓~~C~~AGGGTAATATACGGCCGCCAC…CCC

and we observe this repair product:

CCC…AAGCTGTCCTTCCCCGGCTAGGGGTAATATACGGCCGCCAC…CCC

How we classify this event depends on how we align the product with the substrate. Here is one plausible alignment for this event:

```
CCC…AAGCTGTCCTTCCCCGGCTAGGGATAA
|||…||||||||||||||||||||||| |||
CCC…AAGCTGTCCTTCCCCGGCTAGGGGTAATATACGGCCGCCAC…CCC
                           |||||||||||||||||||…|||
                     ATAACAGGGTAATATACGGCCGCCAC…CCC
```

In this diagram, the middle sequence is the repair product and the upper and lower ones are the substrate. The alignment has a 6-base microhomology which is imperfect[6] on the left side and a 4-base deletion on the right side.

However, the same event could plausibly be aligned in other ways. Such as …

```
CCC…AAGCTGTCCTTCCCCGGCTAGGGATAA
|||…|||||||||||||||||||||||
CCC…AAGCTGTCCTTCCCCGGCTAGGGGTAATATACGGCCGCCAC…CCC
                           |||||||||||||||||||…|||
                     ATAACAGGGTAATATACGGCCGCCAC…CCC
```

Now there is only a 2-base microhomology and deletions of 4 and 5 bases on the left and right sides.

So which alignment is the right one? We cannot know the answer for any single event. The best we can do is look at the statistical properties of a collection of events obtained under similar circumstances and compare them to another collection where our experimental variable has been modified.

For this comparison to be meaningful, the classification process must be objective, repeatable, and produce plausible alignments.

The Other Problem

The random nature of the end-joining process means one must classify a large number of NHEJ events in order to detect subtle differences. However, classifying hundreds or even thousands of NHEJ events manually can be extremely time-consuming even if it could be done objectively.

A Solution

We designed an algorithm to automate the aligning and classifying of NHEJ events. You provide it with:

1. The original sequence.
2. The position and overhang of the DSB.
3. The sequence(s) of one or more repair products.

The result is a reconstruction of the repair event for each product sequence.

This algorithm was implemented in a program, *NJoiner*, available on request. It can handle even very large collections of repair product sequences.

There are seven parameters critical to the success of this algorithm in reconstructing the event. We made use of a database of 1337 NHEJ sequences generated in our lab plus several other sequences gleaned from published data[7] in order to fine-tune the values of these parameters.

Outline of the Algorithm

Step 1: The original (substrate) sequence is divided according to the DSB. If there is any overhang, those bases are included in both parts. We'll refer to these parts as *seqR* and *seqL* and the product sequence will be *seqP*.

Step 2: The Smith-Waterman-Gotoh algorithm[1,2] is used to produce two alignments: *seqR-seqP* and *seqL-seqP*. These alignments are *local* and use parameters specifically tuned to yield plausible NHEJ reconstructions.

Step 3: Any bases included in both local alignments are designated *microhomology*. Any bases of *SeqL* or *SeqR* lying to the right or left (respectively) of their local alignments are designated deletions. Any bases of the product sequence lying between the two local alignments is considered to be an insertion (also called "filler").

Step 4: The two local alignments plus the various designations made in Step 3 are combined in such a way that any matched bases in either alignment are included in the combined alignment. This inclusion might necessitate the introduction of additional gaps.

Step 5: If any long stretch of filler is identified in Step 3, another Smith-Waterman[1] alignment process is used to find any possible template within the original sequence.

Step 6: The resulting reconstruction is used to generate a text diagram showing all features of the proposed NHEJ event. For example, the output below shows deletions of 4 bases from the left side and 8 from the right. It also shows 8 bases of filler, 7 of which were identified as templated in direct orientation from a point a few bases to the right of the cut site.

```
CCC…CTTCCCCGGCTAGGGATAA
|||…|||||||||||||||
CCC…CTTCCCCGGCTAGGGGAATATACGTAATATACG…CCC
                           ||||||||||…|||
                     ATAACAGGGTAATATACG…CCC
```

Filler template:
CCC…CTTCCCCGGCTAGGG^ATAA^CAGGGTAATATACGGCC…CCC

Cool Features of NJoiner

1. It can take input from a directory of ".ab1" files as generated by Applied Biosystems™ equipment. Alternatively, it can accept sequence data from a text file or the clipboard.
2. *NJoiner* uses multithreading. If your computer has several cores, this feature allows it to process sequences much faster. On a typical desktop Mac, it can analyze about 1000 NHEJ sequences per second.
3. All parameters are readily adjusted in preference panels making it easy to tinker with the parameters if the default values aren't optimal for a given data set. You can also specify the fonts, colors, etc., used in the text diagrams.
4. In addition to the text diagrams, output is also provided in tabular form to allow statistical analysis of any feature of interest. This output can be used, for example, to identify specific recurring NHEJ events.
5. *NJoiner* is free on request.

Limitations of NJoiner

1. *NJoiner* runs only on MacOS X 10.5 or later.
2. The substrate is assumed to be a single DSB. *NJoiner* cannot handle experimental designs where two cut sites are present.
3. *NJoiner* does not handle certain aberrant NHEJ events. Specifically, triplications or very long deletions extending beyond the given original sequence are not properly reconstructed.

NJoiner Analysis of a Large Database of NHEJ Events

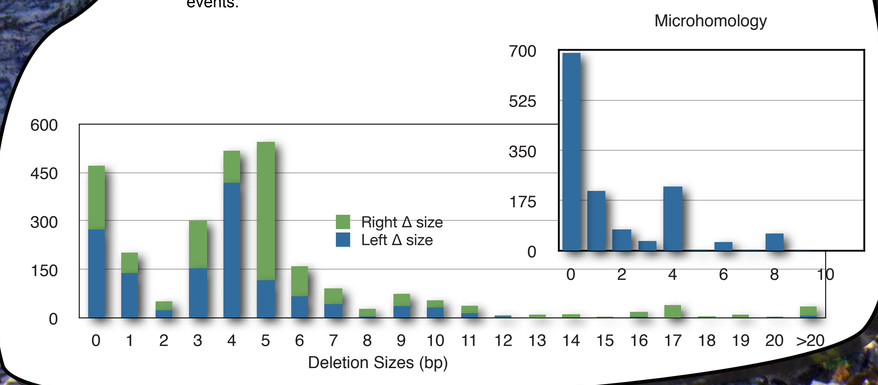
We assembled a collection of NHEJ events from various experiments to use to refine and test the *NJoiner* algorithm. After removing those with poor quality sequence, large deletions, or other problems, we were left with 1337 end-joined sequences for which *NJoiner* produced a plausible reconstruction. By a trial-and-error process, we optimized the parameters of the Smith-Waterman-Gotoh[1,2] alignment algorithm to obtain the following default set:

Match:	4	Deletion Open:	-16
Mismatch:	-8	Deletion Extend:	-4
Insertion Open:	-12	Ambiguous Match:	0
Insertion Extend:	-4		

With these settings, we obtain the distributions shown below for deletion sizes and microhomology lengths. About half of the sequences (647) had some microhomology, averaging 3.26 bp in length. Of these, more than one-third (113) were imperfect on at least one side.

As an example of the effect of changing the parameters, if one makes the mismatch penalty more severe by changing it from -8 to -12, the number of events with microhomology decreases slightly from 647 to 631 while the number where the microhomology was imperfect decreases to 84. The average length of microhomology remains about the same (3.28 bp).

Because this collection was gleaned from several experiments with differences in variables such as organism's age and genomic position, these numbers should not be thought of as a sample of "normal" NHEJ events. However, they do serve to illustrate the utility of the *NJoiner* algorithm for analyzing a large collection of events.



References

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